

Take a simple idea and take it seriously.

Charlie Munger

quoteofand

Agenda

- ① Spark History server
- ② Tabs in SHS
- ③ Important metrics in SHS
- ④ OOM - An overview
- ⑤ Analyse OOM

Sherlock Holmes

Investigate the history

Loading Data - DF

- ① schema already available → no inference
no job
- ② schema should be inferred → sample data
to get schema

Job:

- ① when you perform action

Stage

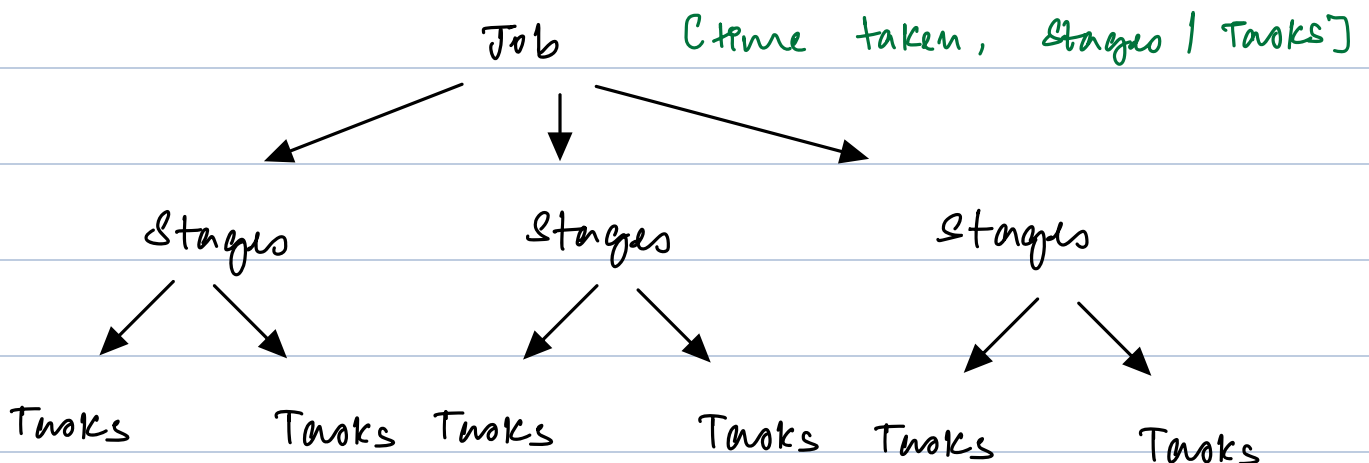
- ① A new stage is created if you want to execute a wide transformation

shuffling of data

Task

- ① smallest unit of work to be performed

Hierarchy



Causes of OOM:

Driver OOM:

- ① Broadcasting large DF
- ② The coordinating process runs out of memory

DF.collect()

DF.toPandas()

one error message

- ③ smaller partitions

java.lang.OutOfMemoryError

Executor OOM:

- ① Data skewed
- ② Incorrectly sized Executor memory
- ③ Unoptimized wide transformation
- ④ caching Big DF or RDD's
- ⑤ Memory intensive UDF